Question 1: A, B and C take turns flipping a coin. The first one to get a head wins.

I Define the sample space.

II Define the following events

1) A wins,
2) B wins,
3) Neither A nor B win

Question 2: Let E, F and G be sets in a sample space. Prove the following relations:

I $E \cap F \subseteq E \subseteq E \cup F$.

II If $E \subset F$ then $F^c \subset E^c$.

III $F = (F \cap E) \cup (F \cap E^c)$.

Question 3: Let E, F and G be sets in a sample space. Prove the following relations:

I $P[E \cup F \cup G] = P[E] + P[F] + P[G] - P[E^c \cap F \cap G] - P[E \cap F^c \cap G] - P[E \cap F \cap G^c] - 2P[E \cap F \cap G]$.

II $P[E \cap F^c] = P[E] - P[E \cap F]$.

Question 4: For a finite set A, let $|A|$ denote the number of elements in A. Show that $|A \cup B| = |A| + |B| - |A \cap B|$.

Question 5: Show that the probability of exactly one of the events E or F occurs equals $P[E] + P[F] - 2P[E \cap F]$.

Question 6: In general show that:

I $P[E \cap F] \geq P[E] + P[F] - 1$.

II $P[E_1 \cap \cdots E_n] \geq P[E_1] + \cdots P[E_n] - (n-1)$.

Question 7: A random experiment consists of selecting two balls in succession from an urn containing two black balls and one white ball.

1) Specify the sample space for this experiment
2) Suppose that the experiment is modified that the ball is immediately put back into the urn after the first selection. What is the sample space now?

Question 8: A random experiment consists of selecting two balls in succession from an urn containing two black balls and one white ball.

1) What is the relative frequency of the outcome (white, white) in a large number of repetitions of the experiment?

2) Does the outcome of the second draw from the urn depend in any way on the outcome of the first draw in this experiment?

Question 9: Let E, F and G be sets in sample space. Prove

$$P[E \cup F \cup G] = P[E]+P[F]+P[G]-P[E^c \cap F \cap G]-P[E \cap F^c \cap G]-P[E \cap F \cap G^c]-2P[E \cap F \cap G].$$

Question 10:

1) How many different 7-place license plates are possible if the 2 first places are letters and the other 5 are decimal numbers?
2) Suppose now that no letter or number can be repeated in a single license plate, how many different license plates are possible?

Question 11: Suppose that A and B are mutually exclusive events for which $P[A] = 0.3$ and $P[B] = 0.5$. What is the probability that:

1) Either A or B occurs;
2) Both A and B occur.

Question 12: A total of 8 percent of Canadians smoke cigarettes, 3 percent smoke cigars, and 2 percent smoke both cigars and cigarettes.

1) What percentage of Canadians smoke neither cigars or cigarettes?
2) What percentage smoke cigars but not cigarettes?